

2654. Minimum Number of Operations to Make All Array Elements Equal to 1

Solved ●

Medium

Topics



Hint

You are given a **0-indexed** array `nums` consisting of **positive** integers. You can do the following operation on the array **any** number of times:

- Select an index `i` such that $0 \leq i < n - 1$ and replace either of `nums[i]` or `nums[i+1]` with their gcd value.

Return the **minimum** number of operations to make all elements of `nums` equal to `1`. If it is impossible, return `-1`.

The gcd of two integers is the greatest common divisor of the two integers.

Example 1:

Input: `nums = [2,6,3,4]`**Output:** 4**Explanation:** We can do the following operations:

- Choose index `i = 2` and replace `nums[2]` with $\text{gcd}(3,4) = 1$. Now we have `nums = [2,6,1,4]`.
- Choose index `i = 1` and replace `nums[1]` with $\text{gcd}(6,1) = 1$. Now we have `nums = [2,1,1,4]`.
- Choose index `i = 0` and replace `nums[0]` with $\text{gcd}(2,1) = 1$. Now we have `nums = [1,1,1,4]`.
- Choose index `i = 2` and replace `nums[3]` with $\text{gcd}(1,4) = 1$. Now we have `nums = [1,1,1,1]`.

Example 2:

Input: `nums = [2,10,6,14]`**Output:** -1**Explanation:** It can be shown that it is impossible to make all the elements equal to 1.

Constraints:

- $2 \leq \text{nums.length} \leq 50$
- $1 \leq \text{nums}[i] \leq 10^6$

Seen this question in a real interview before? 1/5

Yes No

Accepted **82,918** / 151.8K | Acceptance Rate **54.6%**

Topics



Hint 1



Hint 2



Discussion (90)

